

🔍 XAI: Input Gradients & Privacy Risks

FAMAF - UNC

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eXplainable
Artificial Intelligence

Disclaimer ---

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

Spring 2023, Harvard University

<https://interpretable-ml-class.github.io/>

🍷 Paper 1: Do Input Gradients Highlight Discriminative Features? ---



Do Attribution Methods Work? ---

Central Question

Do input gradient attribution methods actually highlight features that drive model predictions?

- Attribution methods assign **importance scores** to input features
- Common assumption: higher gradient magnitude $\Rightarrow$ higher feature contribution
- But... how do we *rigorously evaluate* this?

Challenge

Existing evaluations are often qualitative or rely on synthetic benchmarks — lacking a principled framework

🍷 Assumption (A) ---



🍷 Key Contributions ---

DiffROAR: a new evaluation framework

Systematically evaluate whether attribution methods highlight truly discriminative features via retrain-and-compare

- **Multiple benchmarks:** SVHN, FashionMNIST, CIFAR-10, ImageNet-10
- **BlockMNIST:** controlled dataset isolating discriminative vs. spurious features
- **Feature Leakage Hypothesis:** theoretical explanation for when attribution *appears* to work
- **Theorem:** formal characterization of feature leakage conditions

(Shah et al. 2021)

Related Work ---

****Sanity Checks**** (Adebayo et al.)

- Randomize model weights or labels - Attribution must change accordingly

****Fidelity-based Evaluation****

- Mask top features, measure accuracy drop - Single-pass (no retraining)

****Adversarial Robustness****

- Attribution fragility under adversarial perturbations

Limitation of Prior Work

No evaluation establishes whether top-attributed features are truly *discriminative* for model predictions

DiffROAR Framework ---

🍷 Setting ---



🍷 Unmasking Schemes ---



🍷 Predictive Power & DiffROAR Metric ---

Predictive Power (PredPower_k): test accuracy of a model retrained with only the top- k (or bot- k) features preserved

Definition

$$\text{DiffROAR} = \text{PredPower}(\text{top-}k) - \text{PredPower}(\text{bot-}k)$$



Experiment 1: Image Classification ---

🍷 Experimental Setup ---



🍷 Results: Benchmark Datasets ---



🍷 Experiment 2: BlockMNIST ---

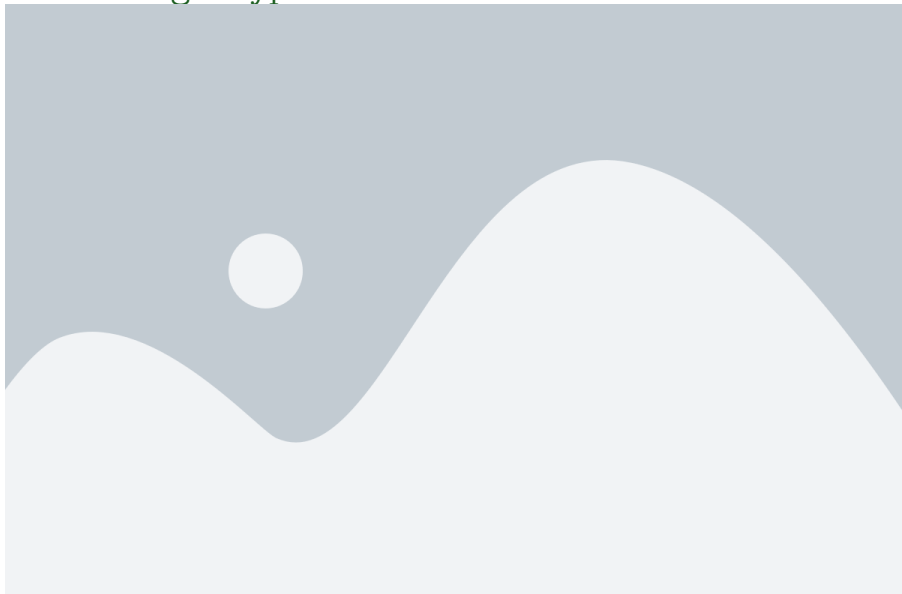
🍷 BlockMNIST Dataset ---



🍷 "Not Always!" ---

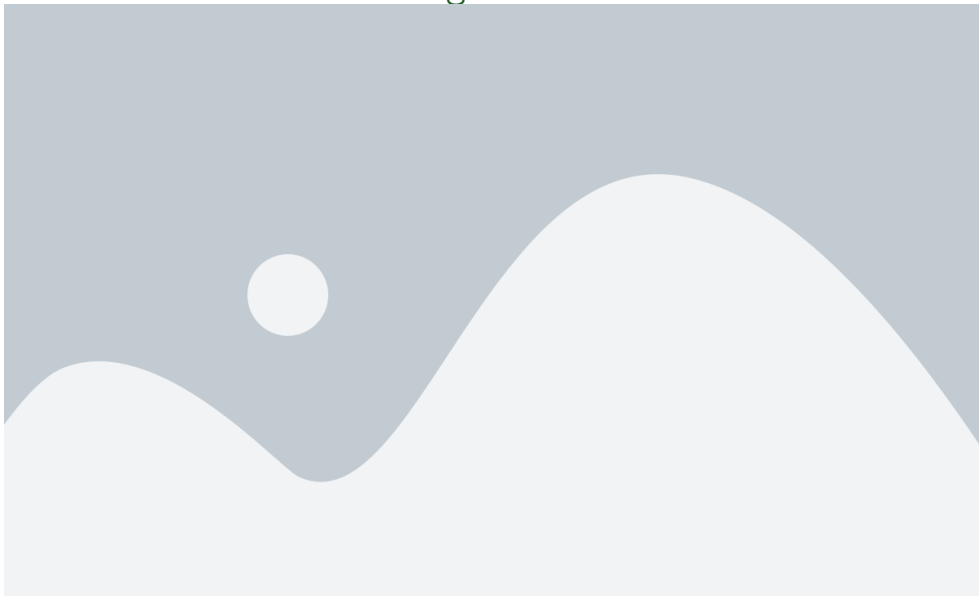


🍷 Feature Leakage Hypothesis ---



🍷 Theoretical Analysis: Feature Leakage ---

🍷 Theorem 1: Feature Leakage ---



🍷 Empirical Validation ---



🍲 Conclusions: DiffROAR ---

Negative Result

Input gradient methods do **not** reliably highlight discriminative features on standard image benchmarks

DiffROAR ≈ 0 for most methods on SVHN, FashionMNIST, CIFAR-10, ImageNet-10

Key Nuance

Feature leakage explains apparent successes:

- BlockMNIST works due to spatial separation
- Not evidence of general semantic understanding

Discussion Questions: DiffROAR ---

- ① DiffROAR requires **retraining** — is this a realistic evaluation paradigm in practice?
- ② Does the feature leakage hypothesis generalize beyond synthetic datasets?
- ③ Can attribution methods be designed to be provably robust to feature leakage?
- ④ If attribution methods don't highlight discriminative features, what are they actually highlighting?

🍷 Paper 2: On the Privacy Risks of Algorithmic Recourse ---



🍷 Privacy Risks in XAI ---

Central Question

Can XAI mechanisms — specifically **algorithmic recourse** — leak private information about training data?



Algorithmic Recourse ---

Context: a rejected loan applicant receives actionable steps to get approved

$$x' = \arg \min_{x' \in \mathcal{A}^P} \ell(f_\theta(x'), 1) + \lambda \cdot c(x, x')$$

- x' : counterfactual (the recourse)
- $c(x, x')$: cost/distance from x to x'
- $\mathcal{A}^P$: set of actionable features

Definition

CFD (Counterfactual Distance): $c(x, x')$ — distance from input x to its recourse x'

Key Observation

The CFD output by recourse algorithms may reveal whether x was in the training set

🍷 Previous Works ---



****Model Extraction via Counterfactuals**** -

Novel Contribution

First **single-query** membership inference (MI) attack using counterfactual distances

- ① **CFD Thresholding Attack**: simple threshold on counterfactual distance
- ② **CFD LRT Attack**: likelihood ratio test using shadow models (more powerful)
- ③ **Theoretical bounds**: DP-based upper bounds on adversary success
- ④ **Empirical evaluation**: diverse datasets (Adult, HELOC, Diabetes) and recourse methods (SCFE, GS, CCHVAE)

(Pawelczyk, Lakkaraju, and Neel 2022)

🍷 MI Attacks: Background ---

Classic Membership Inference:

- M_{Loss} : MEMBER if $\ell(f(x), y) < \tau$ (training samples have lower loss)
- **Loss LRT** (Carlini et al.): likelihood ratio test using shadow models — more powerful



Recourse-based MI Game ---

Owner $\mathcal{O}$ (model provider):

- ① Trains f_θ on training set S
- ② Provides recourse $\mathcal{R}(x) \rightarrow x'$ when queried

Adversary $\mathcal{A}$ (attacker):

- ① Queries $\mathcal{R}(x)$ for target sample x
- ② Receives $\text{CFD} = c(x, x')$
- ③ Predicts: is $x \in S$?

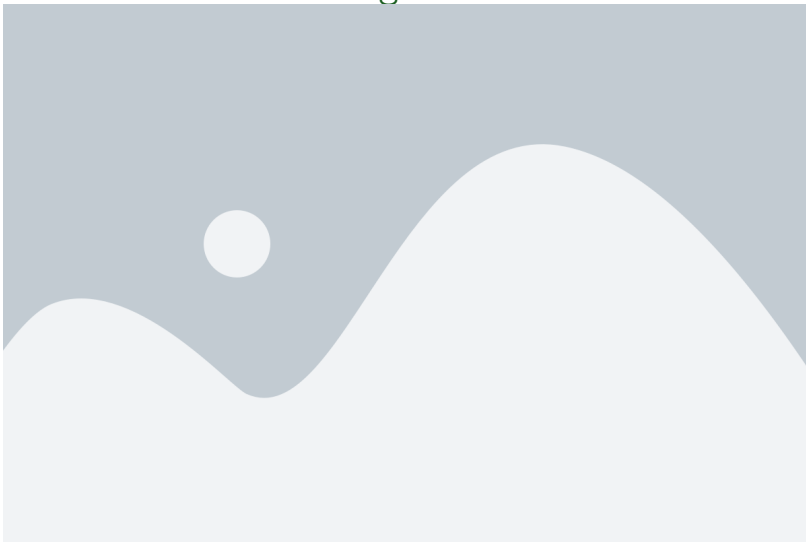
Single-Query Constraint

The adversary can only query the recourse system **once** per sample — making this a realistic threat model

🍷 Intuition: Why CFD Works ---



🍷 Attack 1: CFD Thresholding ---



🍷 Attack 2: CFD Likelihood Ratio Test ---



🍷 Privacy Bounds via Differential Privacy ---



Experimental Setup ---

****Datasets:****

- Adult (income prediction) - HELOC (credit risk) - Diabetes - Synthetic (controlled)

****Recourse Methods:****

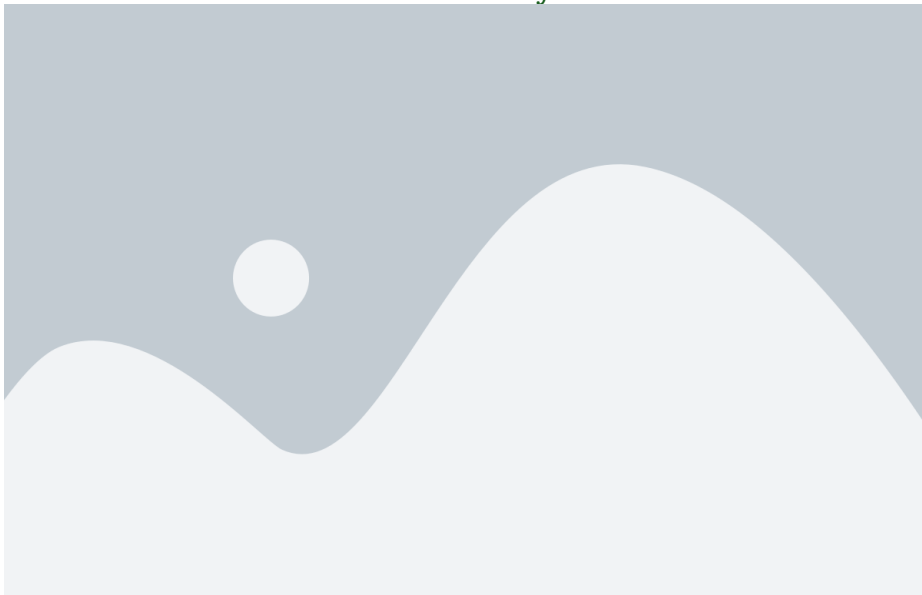
- SCFE - GS (Growing Spheres) - CCHVAE

****Baselines:**** M_{Loss} threshold, Loss LRT

🍷 Attack Efficiency: ROC Curves ---



🍷 Effect of Feature Dimensionality ---



🍷 Effect of Model Architecture ---



🍷 Conclusion: Privacy Risks ---

****Novel Attacks****

- Leverage recourses to infer training data membership - MI attacks using counterfactual distances (****CFD****) - **Single-query** — practical threat model

****Evidence of Privacy Leakage****

- Recourse algorithms carry membership signal
- **Explainability-privacy tradeoff** is real - Attacks effective across diverse domains (lending, healthcare, law)

Limitations ---

- CFD is only a **heuristic** — an approximation of distance to the decision boundary
- Single-query assumption (adversary can only query once per sample)
- Must assume adversary knows the optimal threshold maximizing TPR at fixed FPR
- Paper highlights the **problem**, not yet a **solution**
- Evaluated on binary classification tasks only

Future Work ---

- **Generalization**: can recourse lead to reconstruction attacks or attacks on training data statistics?
- **Other XAI mechanisms**: which other explanation methods involve privacy violations?
- **Solutions to protect privacy**: train models that provide recourse while mitigating privacy risks
 - ▶ How to construct faithful explanations that don't leak training data?
 - ▶ What is the privacy-utility trade-off?

(Pawelczyk, Lakkaraju, and Neel 2022)

Discussion Questions: Privacy Risks ---

- ➊ Given the explainability-privacy tradeoff, what is the role of **ML practitioners** vs. **end users** in setting explainability and privacy benchmarks?
- ➋ Both privacy and explainability cultivate user *trust* in ML models. In what situations would you prioritize one pillar over the other?
- ➌ Besides recourse, what other XAI mechanisms might lead to privacy violations?
- ➍ Is it even possible to have **private explanations**? Is this worth pursuing?

Thank You!

References --- |

- Pawelczyk, Martin, Himabindu Lakkaraju, and Seth Neel. 2022. "On the Privacy Risks of Algorithmic Recourse." In *Advances in Neural Information Processing Systems*.
- Shah, Harshay, Kaustav Tamuly, Prateek Jain, and Praneeth Netrapalli. 2021. "Do Input Gradients Highlight Discriminative Features?" In *Advances in Neural Information Processing Systems*.